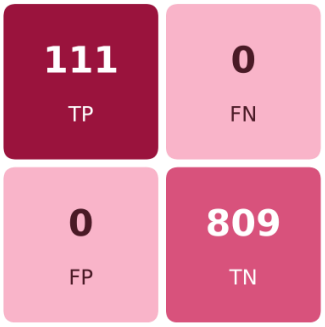
PART H · Q26

Final Analysis & Reporting

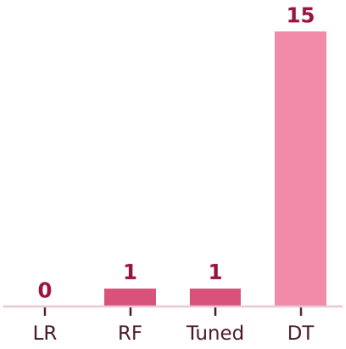
Risk Alert Classifier



Class balance



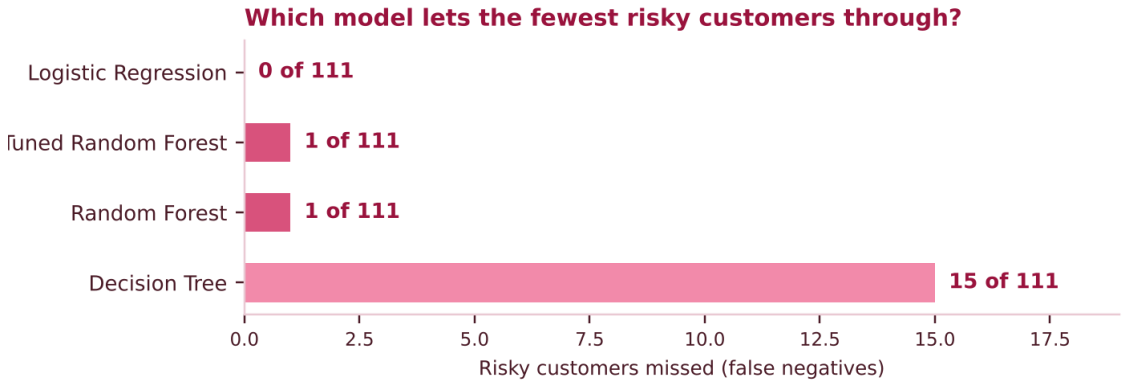
Final model: LR



Risky customers missed

1. Best Classification Model & Justification

The final model is **Logistic Regression**. On the 920-row test set it produced **0 false negatives and 0 false positives** (accuracy, precision, recall, F1 and AUC all 1.0), catching all 111 risky customers.



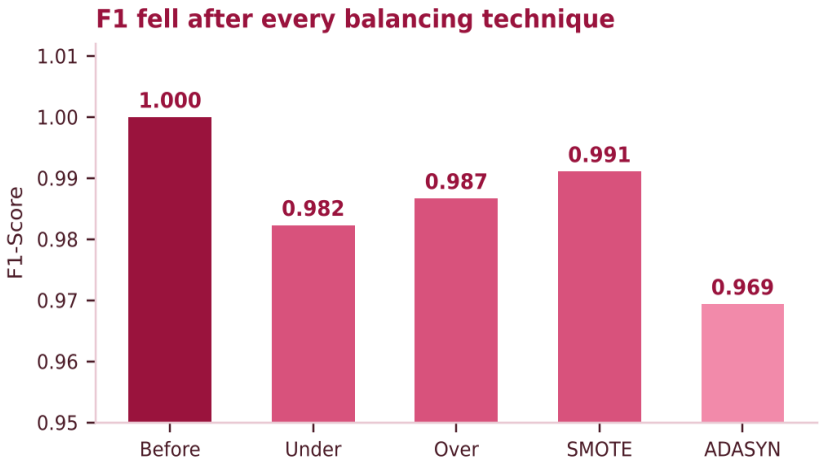
Random Forest (test accuracy 0.9967, tuned 0.9978) missed 1 risky customer, and the Decision Tree missed 15 because a single tree overfits (train 1.0 vs test 0.9707). Logistic Regression is also the easiest to explain to business teams, which matters for a risk tool.

Caveats: customer_id was accidentally used as a numeric feature and should be dropped; the 0-vs-1 missed-customer gap rests on 111 risky cases from a single split, so cross-validation is advised before deployment.

2. Impact of Imbalance Handling Techniques

Only **12.11%** of customers are risky (557 of 4,600). Under-sampling, over-sampling, SMOTE and ADASYN were applied to the training data only. Recall stayed at 1.0 in every case, but **F1 dropped** because each technique added a few false alarms.

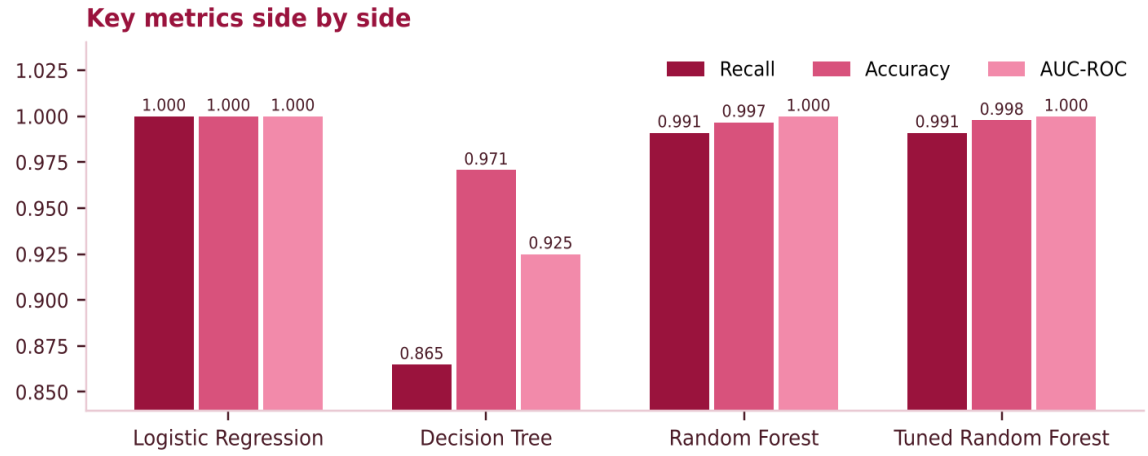




The baseline was already perfect, so balancing had nothing to fix. SMOTE hurt least (F1 0.991) and ADASYN most (0.969). Balancing becomes useful when the model actually misses minority cases, which was not the case here.

3. Comparison of Performance Metrics

Model	Accuracy	Recall	AUC-ROC	Risky missed
Logistic Regression	1.0000	1.0000	1.0000	0 of 111
Decision Tree	0.9707	0.8649	0.9250	15 of 111
Random Forest	0.9967	0.9910	0.9999	1 of 111
Tuned Random Forest	0.9978	0.9910	0.9999	1 of 111



Tuning raised Random Forest F1 from 0.9865 to 0.9910 (best: 150 trees, depth 15, min split 10). Tuned Random Forest is a strong backup, but only Logistic Regression reached zero missed customers.

4. Business Interpretation of False Positives & False Negatives

	False Negative (Type II)	False Positive (Type I)
Meaning	Risky customer predicted as safe	Safe customer predicted as risky
Business cost	Defaults, direct financial loss, fraud goes unnoticed	Lost business, annoyed customers, wasted review effort
Severity	High: usually the costlier error	Moderate: recoverable with manual review

Because a missed risky customer costs more than an unnecessary alert, **recall should be prioritised**. Logistic Regression gives full recall with no false alarms, so it fits this need best.

- **Recommendation:** deploy Logistic Regression, remove customer_id, and validate with cross-validation.
- **Monitoring:** track recall monthly and retrain if risky cases start slipping through.